TEU00311

What is the Internet doing to me? (witidtm)

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https://github.com/sftcd/witidtm
https://down.dsg.cs.tcd.ie/witidtm

WebRTC – making calls on the web

WebRTC is how many online calls work

- WebRTC == Web Real Time Communications
- WebRTC kinda saved us during the pandemic and was finished just in time
- Pretty much any service that does calls does WebRTC at least for interop with callers who are just using a browser
  - That includes whatsapp, signal calls and similar
  - Some clients/services may deviate from WebRTC standards in some cases, for go-faster stripes or added functionality

How WebRTC works

- It’s a complicated beast, complications arising because of:
  - NAT (network address translation)
  - Audio/Video codec variations and needing to pick the same options
  - Security
  - Interoperability with legacy VoIP technologies
  - Layering of protocols that handle the above
- Details from a 2017 presentation:
  - https://datatracker.ietf.org/meeting/100/materials/slides-100-edu-sessm-webrtc-tutorial-part-1-00
  - We’ll pop over for a few of those slides...

WebRTC Privacy

- Audio/video permissions in your browser are “sticky” so if you allow a web site to use those, you might leave that permission set forever, potentially allowing that site to eavesdrop on you anytime you visit
  - Browsers have config options that allow you to go back and turn that off for servers
- The way WebRTC figures out how to setup calls can expose your IP address even if you’re using a VPN
  - Some browsers have a config setting to avoid this, but usually with some incomprehensible name, e.g. “WebRTC IP handling policy” in Brave
  - https://support.brave.app/hc/en-us/articles/360017989132-How-do-I-change-my-Privacy-Settings#webrtc
  - Mobile apps that do calling probably don’t have that option
- I have no clue if trying to set/disguise your background is useful;-)
- Adding an “AI bot” likely breaches all callers’ privacy
- Zoombombing – if a call URL is all you need, someone else may join for fun

WebRTC Security

- Your call (audio/video/chat) data is encrypted in transit but may not be properly authenticated
- Peer-to-peer/1:1 calls may well be end-to-end encrypted (browser to browser), but the server could cheat on the setup
- For multiparty calls, usually a server on the call sees the plaintext
  - Different codecs for different call participants, e.g. some on phone, some on laptop
  - Chat logs, recording, translation etc. need plaintext
- Good paper on WebRTC security: Bach et. al: “Analyzing the WebRTC Ecosystem and Breaking Authentication in DTLS-SRTP”, Usenix 2026
  - https://www.usenix.org/system/files/usenixsecurity26-bach.pdf

WebRTC Usage

- I think…
  - Fully-online conferences, and “family-chats” with video, are crap
  - If you already know the people on a call, then doing business this way can be quite effective, esp if you meet them face-to-face at least annually
  - Even when in the room, the chat side-channel can be very useful
  - Recording lectures turns out counter-productive for a cohort as students plan to, but don’t really, go back and review
- What do you think?
  - Of online lectures?
  - Of online calls generally, for work or personal use?
  - What’d you change/add/remove?